

Dilman Sandhu

✉ dilman3349@gmail.com | 📞 +91 83024-80312 | 🌐 dilman-sandhu | 📱 dilman0812

Summary

Software Engineering student with hands-on experience in backend-oriented development using Java, Node, and SQL. Strong in debugging workflows, validating system behavior, analyzing logs, and ensuring data consistency across application layers. Comfortable working with backend logic, manual API testing, and structured problem-solving. Seeking entry-level backend or software engineering roles where reliability, clarity, and ownership matter.

Skills

- **Programming Languages:** Core Java, JavaScript (Node.js), Python, SQL
- **Backend Development:** Backend logic implementation, RESTful API development concepts, request-response lifecycle, input validation, error handling, and workflow debugging
- **Databases & Persistence:** MySQL, transaction handling, basic data consistency concepts
- **API Testing & Debugging:** Manual API testing using Postman (basic), validating API responses, analyzing logs, and verifying database state changes
- **Developer Tools & Environment:** Git, Linux command-line basics, Postman, VS Code, IntelliJ IDEA
- **Software Engineering Concepts:** Object-Oriented Programming, modular code design, debugging techniques, error handling, basic concurrency awareness

Projects

Travel Planner Platform – Backend Workflow Development

Feb 2024 – May 2024

- Designed backend logic supporting booking workflows, user onboarding, and guide assignment.
- Debugged and stabilized **15+ workflow issues** by analyzing logs, tracing request flows, and identifying inconsistent state transitions.
- Tested and validated **20+ API endpoints** for response correctness, error handling, and database synchronization.
- Maintained structured test logs and documentation to track recurring issues and improve system reliability.

Core Banking System & ATM Simulator – Backend Transaction System

Aug 2023 – Oct 2023

- Built a Java-based banking system supporting account creation, authentication, deposits, withdrawals, and transaction logs using JDBC and MySQL.
- Verified **100+ transaction scenarios** including invalid PINs, insufficient balance, and concurrency cases.
- Improved system stability by debugging SQL queries, refining validation logic, and resolving edge cases involving inconsistent state updates.
- Documented test scenarios and expected outcomes to support repeatable backend validation.

Lung Cancer Prediction System – Data Processing Backend Pipeline

Jul 2024 – Mar 2025

- Processed and validated **300+ CT scan inputs** while troubleshooting preprocessing and inference inconsistencies.
- Reduced noisy or unusable input cases by **20%** through improved filtering and preprocessing logic.
- Developed modular Python components following a clean input → processing → output architecture.
- Documented pipeline behavior and test results to support reproducibility and debugging.

Achievements

- Created structured validation workflows and documentation across projects, improving consistency in debugging and analysis.
- Supported onboarding and operations for **6,000+ students**, demonstrating organization and leadership.
- Organized campus technical and sports events with **3,000+ participants**.

Education

Vellore Institute of Technology

2022 – 2026

B.Tech in Computer Science Engineering, CGPA: 8.2/10